```
##A-1.1/A-3.4:Python Programme to do Addition, Subtraction, Multiplication, Integer Divisi
def add():
    summation = 0
    arr = []
    ele_num = int(input("Enter total number of elements you want to add:"))
    for i in range(0, ele_num):
        ele = int(input("Enter element: "))
        arr.append(ele)
        summation = summation + arr[i]
    return summation
def sub():
    subtract = 0
    a = int(input("Enter First element: "))
    b = int(input("Enter Second element: "))
    subtract = a-b
    return subtract
def mul():
    multiply = 0
    a = int(input("Enter Element 1: "))
    b = int(input("Enter Element 2: "))
    multiply = a*b
    return multiply
def int_div():
    integer_div = 0
    a = int(input("Enter Element 1: "))
    b = int(input("Enter Element 2: "))
    integer_div = a//b
    return integer_div
def float_div():
    f_d = 0
    a = float(input("Enter Element 1: "))
    b = float(input("Enter Element 2: "))
    f_d = a/b
    return f_d
def exp():
    expo = 0
    a = float(input("Enter Base element: "))
    b = int(input("Enter Power element: "))
    expo = a**b
    return expo
def mod_div():
    m_d = 0
    a = int(input("Enter First element: "))
    b = int(input("Enter Second element: "))
    m_d = a%b
    return m_d
def avg():
    summation =0
    ave = 0
    arr = []
    ele_num = int(input("Enter total number of elements you want to add:"))
    for i in range(0, ele_num):
        ele = int(input("Enter element: "))
        arr.append(ele)
        summation = summation + arr[i]
    ave = summation/ele_num
    return ave
print("List of Choice:\n1. Addition\n2. Subtraction\n3. Multiplication\n4. Integer Divisio
```

```
cho = int(input("Enter your choice: "))
match cho:
    case 1:
        Addition = add()
        print("Result of the Summation of the elements is: ", Addition)
    case 2:
        Subtraction = sub()
        print("Result of Subtraction of the elements is:", Subtraction)
    case 3:
        Multiplication = mul()
        print("Result of the Multiplication of the elements is: ", Multiplication)
    case 4:
        Integer_Division = int_div()
        print("Result of the Integer division of the element is: ", Integer_Division)
    case 5:
        Float_Division = float_div()
        print("Result of the Float division of the elements is: ", Float_Division)
    case 6:
        Exponentiation = exp()
        print("Result of the Exponentiation of the elements is: ", Exponentiation)
    case 7:
        Modulo_Division = mod_div()
        print("Result of the Modulo division of the elements is: ", Modulo_Division)
    case 8:
        Average_of_Numbers = avg()
        print("Result of the Average of the elements is: ", Average_of_Numbers)
    case 9:
        exit()
    case default:
        print("Wrong Choice Entered.")
```

```
List of Choice:
1. Addition
2. Subtraction
3. Multiplication
4. Integer Division
5. Float Division
6. Exponentiation
7. Modulo Division
8. Average of Numbers
9. Exit Operation
Enter your choice: 9
```

```
#A-1.2: Write a Python program for finding the area of a Circle and Rectangle
def area_cir():
    a_c = 0
    r = float(input("Enter the radius of the Circle: "))
    a_c = 22/7*(r**2)
    return a_c
def area_tri():
    a_t = 0
    h = float(input("Enter Height of the Triangle: "))
    b = float(input("Enter Base of the Triangle: "))
    a_t = 1/2*h*b
    return a_t
print("List of Choice:\n1. Area Of Circle\n2. Area Of Triangle")
print("Enter your choice")
cho = int(input("Enter your Choice: "))
match cho:
```

```
    case 1:
        Area_Circle = area_cir()
        print("The area given Circle: ", Area_Circle, "Square Unit")
    case 2:
        Area_Triangle = area_tri()
        print("The area given Triangle: ", Area_Triangle, "Square Unit")
```

```
List of Choice:
1. Area Of Circle
2. Area Of Triangle
Enter your choice
Enter your Choice: 1
Enter the radius of the Circle: 4
The area given Circle:  50.285714285714285 Square Unit
```

```
#A-1.3: Write a Python Programme to print the first place of any number taken by user.
num = int(input("Enter any numer you want: "))
rem = 0
temp = 0
while (num >= 10):
    num = num // 10
print("The Ones Place/First Place of a Number is: ",num)
```

```
Enter any numer you want: 34
The Ones Place/First Place of a Number is:  3
```

```
#A-1.4: Write a Python Programme to convert Celcius to Faranheit and vice versa.
def Cel_to_Far():
  C=float(input("Enter temperature in Celcius: "))
  F=(C * 9/5)+32
  return F
def Far_to_Cel():
  F=float(input("Enter temperature in Faranheit: "))
  C=((F-32)*5)/9
  return C
print("List of choices\n1. Celcius to Faranheit\n2. Faranheit to Celcius.\n3. Exit.\n")
choice=int(input("Enter your choice: "))
match choice:
   case 1:
    Faranheit = Cel_to_Far()
    print("The converted temperature in Faranheit: ",Faranheit)
   case 2:
    Celcius = Far_to_Cel()
    print("The converted temperature in Celcius: ",Celcius)
   case 3:
    exit()
```

```
List of choices
1. Celcius to Faranheit
2. Faranheit to Celcius.
3. Exit.

Enter your choice: 2
Enter temperature in Faranheit: 32.0
The converted temperature in Celcius:  0.0
```

```
#A-1.5: WAP in Python to calculate the total amount of money in Piggy Bank.
def Tenrupee():
  CT=int(input("Enter total Numbers of 10 rupee coins: "))
  Amount = CT*10
  return Amount
def Fiverupee():
  CF=int(input("Enter total Numbers of 5 rupee coins: "))
  Amount2 = CF*5
  return Amount2
def Tworupee():
  CTo=int(input("Enter total Numbers of 2 rupee coins: "))
  Amount3 = CTo*2
  return Amount3
def Onerupee():
  CO=int(input("Enter total Numbers of 1 rupee coins: "))
  Amount4 = CO
  return Amount4
Savings_in_PiggyBank = Tenrupee()+Fiverupee()+Tworupee()+Onerupee()
print("The total savings in the Piggy Bank is: ",Savings_in_PiggyBank)
```

```
Enter total Numbers of 10 rupee coins: 10
Enter total Numbers of 5 rupee coins: 5
Enter total Numbers of 2 rupee coins: 15
Enter total Numbers of 1 rupee coins: 20
The total savings in the Piggy Bank is:  175
```

```
#A-1.6: CHECK ODD OR EVEN.
num=int(input("Enter any Number you want to check: "))
if(num%2==0):
  print("The given number ")
  print(num)
  print("is even.")
else:
  print("The given number ")
  print(num)
  print("is odd.")
```

```
Enter any Number you want to check: 46
The given number
46
is even.
```

```
#A-1.7: WAP to find largest number among three numbers.
N1=int(input("Enter the first number to compare: "))
N2=int(input("Enter the second number to compare: "))
N3=int(input("Enter the third number to compare: "))
if(N1>N2 and N1>N3):
    print("The first element ", N1," is greater than others.")
elif(N2>N1 and N2>N3):
  print("The second element ", N2," is greater than others.")
else:
  print("The third element ", N3," is greater than others.")
```

```
Enter the first number to compare: 12
Enter the second number to compare: 45
Enter the third number to compare: 23
The second element  45  is greater than others.
```

```
#A-1.8: WAP to determine the charater entered by the user.
ch = input("Please Enter Your Own Character : ")
if((ch >= 'a' and ch <= 'z') or (ch >= 'A' and ch <= 'Z')):
    print("The Given Character ", ch, "is an Alphabet")
elif(ch >= '0' and ch <= '9'):
    print("The Given Character ", ch, "is a Digit")
else:
    print("The Given Character ", ch, "is a Special Character")
```

```
Please Enter Your Own Character : @
The Given Character  @ is a Special Character
```

```
#A-1.9: WAP to check whether the given alphabet is vowel.
Character = input("Enter any Alphabet: ")
if(Character=='A' or Character=='a' or Character=='E' or Character=='e' or Character=='I'
  print("The given character ",Character," is a Vowel")
else:
  print("The given character ",Character," is a Consonant")
```

```
Enter any Alphabet: s
The given character  s  is a Consonant
```

```
#A-2.3: CHECK LEAP YEAR OR NOT.
year=int(input("Enter any year you want to check: "))
if(year%4==0 and year%100!=0 or year%400==0):
  print("The given year is Leap Year.")
else:
  print("The given year is not Leap Year.")
```

```
Enter any year you want to check: 2000
The given year is Leap Year.
```

```
#A-2.4:Write a Python Code to calculate sum of 'n' natural number.
r=int(input("Enter the range of natural numbers: "))
sum=0
for i in range(0,r+1):
  sum=sum+i
print("The result of the summation of 'n' natural number is:")
print(sum)
```

```
Enter the range of natural numbers: 4
The result of the summation of 'n' natural number is:
10
```

```
#A-2.6:CHECK PALINDROME OR NOT.
num=int(input("Enter any number you want to check Palindrome or not: "))
sum=0
d=0
temp = num
```

```
while(num>0):
  d=num%10
  sum=sum*10+d
  num=num//10
if(sum==temp):
  print("The given Number is Palindrome.")
else:
  print("The given Number is not Palindrome.")
```

```
Enter any number you want to check Palindrome or not: 121
The given Number is Palindrome.
```

```
#A-4.1: WAP to print Tribonacci Series.
range= int(input("Enter Range: "))
n1, n2 = 0, 1
count = 0
if range <= 0:
   print("Please enter a positive integer")
elif range==1:
   print("Fibonacci sequence upto",range,":")
else:
   print("Fibonacci sequence:")
   while count < range:
       print(n1)
       nth=n1+n2+1
       n1=n2
       n2=nth
       count+=1
```

```
Enter Range: 4
Fibonacci sequence:
0
1
2
4
```

```
#A-4.4:Find Factorial of a number using recursion.
num=int(input("Enter the number you want to get Factorial: "))
def fact(num):
  if(num==1 or num==0):
    return 1
  elif(num<0):
    return "Invalid Choice"
  else:
    return num*fact(num-1)
Factorial=fact(num)
print("The factorial of the given number is: ")
print(Factorial)
```

```
Enter the number you want to get Factorial: 5
The factorial of the given number is:
120
```

```
#A-4.5:Print Fibbonacci Series
nterms = int(input("How many terms? "))
n1, n2 = 0, 1
count = 0
```

```
if nterms <= 0:
   print("Please enter a positive integer")
elif nterms == 1:
   print("Fibonacci sequence upto",nterms,":")
   print(n1)
else:
   print("Fibonacci sequence:")
   while count < nterms:
       print(n1)
       nth = n1 + n2
       n1 = n2
       n2 = nth
       count += 1
```

```
How many terms? 8
Fibonacci sequence:
0
1
1
2
3
5
8
13
```

```
#A-5.1: Write a Python Programme to add two numbers using Lambda.
a=int(input("Enter the first number you want to add: "))
b=int(input("Enter the second number you want to add: "))
Sum=lambda a,b:a+b
print("The result of the addition of two numbers is: ",Sum(a,b))
```

```
Enter the first number you want to add: 5
Enter the second number you want to add: 3
The result of the addition of two numbers is:  8
```

```
#A-5.2: Write a Python Programme to find smaller two numbers using Lambda.
a=int(input("Enter the first number: "))
b=int(input("Enter the first number: "))
smaller=lambda a,b:min(a,b)
print("The smaller element is: ",smaller(a,b))
```

```
Enter the first number: 90
Enter the first number: 23
The smaller element is:  23
```

```
#A-5.5: Write a programme that uses split() to split a multiple string using new line.
Initial_String=str(input("Enter any String: "))
New_String=Initial_String.split()
print(New_String)
```

```
Enter any String: I am fine
['I', 'am', 'fine']
```

```
#A-5.6: Write a Python programme to accept a string from user and re-display the same afte
def rem_vowel(string):
```

```
    vowels = ['a','e','i','o','u']
    result = [letter for letter in string if letter.lower() not in vowels]
    result = ''.join(result)
    print(result)
Initial_String=str(input("Enter any String: "))
rem_vowel(Initial_String)
```

```
Enter any String: I am fine
 m fn
```

```
#A-6.1: Write a Python to reverse a string.
Initial_String=str(input("Enter any String: "))
New_String=Initial_String[::-1]
print("The New String is the following:",New_String)
```

```
Enter any String: I am Fine
The New String is the following: eniF ma I
```

```
#A-6.2: Write a programme that uses split() to split a multiple string using new line.
Initial_String=str(input("Enter any String: "))
New_String=Initial_String.split()
print(New_String)
```

```
Enter any String: You are mine
['You', 'are', 'mine']
```

```
#A-6.3: Write a Python programme to accept a string from user and re-display the same afte
def rem_vowel(string):
    vowels = ['a','e','i','o','u']
    result = [letter for letter in string if letter.lower() not in vowels]
    result = ''.join(result)
    print(result)
Initial_String=str(input("Enter any String: "))
rem_vowel(Initial_String)
```

```
Enter any String: go to home
g t hm
```

What is Module?

1. Module allows you to use one or more functions in your programme, even in the programme those functions not have been defined.
2. Module is a file with .py extension, that has definitions of all functions and variables.
3. Modules are generally two type 1. System define or builtin module - which contains builtin functions and variables. 2. User Define Module - it can be define by the user it self depends on the purpose of user functions and variables.

```
# Module(Built-in).
import sys
print(sys.version) # version of your present python system
from math import pi
```

```
print(pi) # pi value from math function
a=int(input("Enter any number: "))
print(pow(a,2)) #pow function.
```

```
3.10.12 (main, Jun 11 2023, 05:26:28) [GCC 11.4.0]
3.141592653589793
Enter any number: 9
81
```

```
#A-7.1: Python programme to print time and calender of a specific month
import datetime
current_time = datetime.datetime.now()
print("The current time of the system is: ",current_time)
import calendar
Year=int(input("Enter year: "))
Month=int(input("Enter month: "))
print("The calender of month ",Month," is:\n")
print(calendar.month(Year,Month))
```

```
The current time of the system is:  2023-08-17 06:41:59.378237
Enter year: 2023
Enter month: 9
The calender of month  9  is:

   September 2023
Mo Tu We Th Fr Sa Su
                1  2  3
 4  5  6  7  8  9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30
```

```
#A-7.2: Python programme to print absolute value, square root and cube of the number.
import math
num = int(input("Enter any number: "))
print("The absolute value of your given number ",num," is: ",abs(num))
print("The squareroot value of your given number ",num," is: ",math.sqrt(num))
print("The cube of your given number ",num," is:",pow(num,3))
```

```
Enter any number: 4
The absolute value of your given number  4  is:  4
The squareroot value of your given number  4  is:  2.0
The cube of your given number  4  is: 64
```

```
#A-7.3(a): Define a factorial function in a user-defined module "MyFact" and find factoria
def fact(int num):
  if(num==1 or num==0):
    return 1
  elif(num<0):
    return "Invalid Choice"
  else:
    return num*fact(num-1)
Factorial=fact(num)
```

```
#A-7.3(b): Define a factorial function in a user-defined module "MyFact" and find factoria
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call d
```

```
#A-7.3(c): Define a factorial function in a user-defined module "MyFact" and find factoria
import sys
sys.path.insert(0,'/content/drive/My Drive')
```

```
#A-7.3(d): Define a factorial function in a user-defined module "MyFact" and find factoria
import MyFact
n=int(input("Enter any number you want to get factorial: "))
Factorial = MyFact.fact(n)
print(Factorial)
```

```
Enter any number you want to get factorial: 5
120
```

```
#A-7.4(a): Define a Large function in a user-defined module "Max" and find max value of th
def large(int N1,int N2,int N3):
  if(N1>N2 and N1>N3):
      print("The first element ", N1," is greater than others.")
  elif(N2>N1 and N2>N3):
    print("The second element ", N2," is greater than others.")
  else:
    print("The third element ", N3," is greater than others.")
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call d
```

```
import sys
sys.path.insert(0,'/content/drive/My Drive')
```

```
#A-7.4(b): Define a Large function in a user-defined module "Max" and find max value of th
import MyFact

num1=int(input("Enter first number: "))
num2=int(input("Enter second number: "))
num3=int(input("Enter third number: "))
print(MyFact.large(num1,num2,num3))
```

```
Enter first number: 2
Enter second number: 5
Enter third number: 1
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
```

```
#A-8.1: Python Programme for interchanging First and Last element.
r=int(input("Enter Range: "))
lst = []
for i in range(0,r+1):3
  x=int(input("Enter Value: "))
  lst.insert(i,x)
  lst[0], lst[-1] = lst[-1], lst[0]
print(lst)
```

```
Enter Range: 3
Enter Value: 12
Enter Value: 4
Enter Value: 45
Enter Value: 67
[67, 12, 4, 45]
```

```
#A-8.2: Python Programme for getting length of array.
r=int(input("Enter Range: "))
lst = []
for i in range(0,r+1):1
  x=int(input("Enter Value: "))
  lst.insert(i,x)
  print(lst)
length=len(lst)
print(length)
```

```
Enter Range: 5
Enter Value: 0
[0]
Enter Value: 1
[0, 1]
Enter Value: 2
[0, 1, 2]
Enter Value: 3
[0, 1, 2, 3]
Enter Value: 4
[0, 1, 2, 3, 4]
Enter Value: 5
[0, 1, 2, 3, 4, 5]
6
```

```
#A-8.3: Python Programme to swap element in list.
r=int(input("Enter Range: "))
lst = []
for i in range(0,r+1):
  x=int(input("Enter Value: "))
  lst.insert(i,x)
  lst[0], lst[-1] = lst[-1], lst[0]
print(lst)
```

```
Enter Range: 3
Enter Value: 12
Enter Value: 34
Enter Value: 45
Enter Value: 456
[456, 12, 34, 45]
```

```
#A-8.4: Python Programme to sort element in list.
r=int(input("Enter Range: "))
lst = []
for i in range(0,r+1):
  x=int(input("Enter Value: "))
  lst.insert(i,x)
  lst.sort()
print("Sorted List is: 9",lst)
```

```
Enter Range: 4
Enter Value: 78
Enter Value: 0
Enter Value: 90
Enter Value: 23
Enter Value: 12
Sorted List is:  [0, 12, 23, 78, 90]
```

```
#A-8.5: Python Programme to swap element in list.
r=int(input("Enter Range: "))
lst = []
for i in range(0,r+1):
  x=int(input("Enter Value: "))
  lst.insert(i,x)
  lst.reverse()
print(lst)
```

```
Enter Range: 3
Enter Value: 12
Enter Value: 23
Enter Value: 34
Enter Value: 45
[45, 23, 12, 34]
```

```
#A-10.1: Write a Python Programme that combine the lists into Dictionary.
String_1=['Name','Age','Salary']
String_2=['Snehasish',18,2500]
New_dictionary=dict(zip(String_1,String_2))
print(New_dictionary)
```

```
{'Name': 'Snehasish', 'Age': 18, 'Salary': 2500}
```

```
#A-10.2: Write a Python Programme to create a Dictionary of Odd Numbers.
r=int(input("Enter your range: "))
even=[]
odd=[]
for i in range(1,r):
    n = int(input('Enter any number: '))
```

```
    if n%2 == 0:
        even.append(n)
    else:
        odd.append(n)
d = {}
d['Even']=even
d['Odd']=odd
print(d)
```

```
Enter your range: 10
Enter any number: 1
Enter any number: 2
Enter any number: 3
Enter any number: 4
Enter any number: 5
Enter any number: 6
Enter any number: 7
Enter any number: 8
Enter any number: 9
{'Even': [2, 4, 6, 8], 'Odd': [1, 3, 5, 7, 9]}
```

```
#A-10.5: Write a Programme to print fibbonaci series using dictionary
Dict={0:0, 1:1}
def fibo(n):
    if n not in Dict:
        val=fibo(n-1)+fibo(n-2)
        Dict[n]=val
    return Dict[n]
n=int(input("Enter the value of n:"))
print("Fibonacci(", n,")= ", fibo(n))
```

```
Enter the value of n:5
Fibonacci( 5 )=  5
```

```
#A-11.Basics: Declaring Class and Object.
class ABC:
  var=10
  def display(self):
    print("Hello Class in py")
obj=ABC()
print(obj.var)
obj.display()
# del.obj
```

```
10
Hello Class in py
```

```
#A-11.Basic: Factorial of number using class.
class FACT:
  def __init__(self,var1):
    self.var1=var1;
    print(var1)
    p=1
    for i in range(1,var1+1):
      p=p*i
    print("The Factorial of the output: ",p)
```

```
var1=int(input("Enter number you want to get factorial: "))
obj=FACT(var1)
```

```
Enter number you want to get factorial: 5
5
The Factorial of the output:  120
```

```
#A.11.Basic: ODD AND EVEN LIST.
class number:
  even=[]
  odd=[]
  def __init__(self,num):
    self.num=num
    if num%2==0:
      number.even.append(num)
    else:
      number.odd.append(num)
n1=number(12)
n2=number(23)
n3=number(34)
n4=number(45)
n5=number(56)
print("Even Number: ",number.even)
print("Odd Number: ",number.odd)
```

```
Even Number:  [12, 34, 56]
Odd Number:  [23, 45]
```

```
#A-11.2: Using a class named circle calculate area, circumference of the circle.
class circle:
  def __init__(self,radius):
    PI = 3.14
    self.radius=radius
    area = PI*(radius**2)
    print("Area of the circle: ",area," squnit")
    circumference = 2*PI*radius
    print("Circumference of the circle: ",circumference," unit")
obj = circle(int(input("Enter radius of the circle: ")))
```

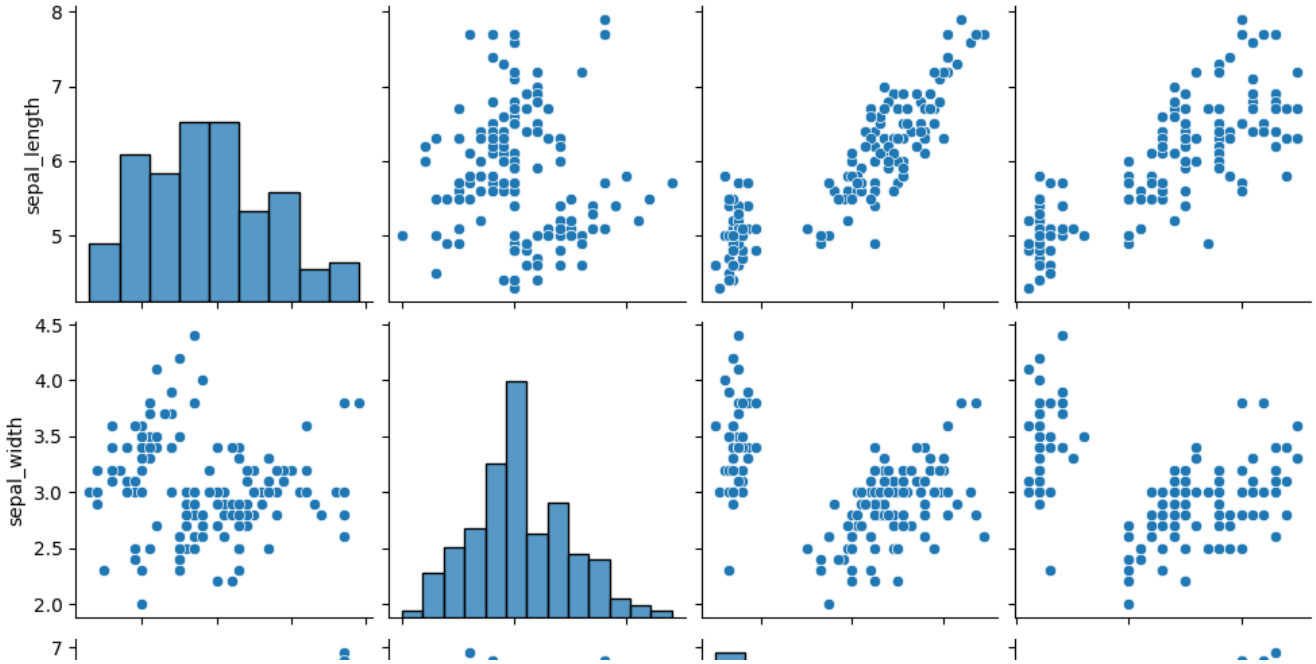
```
Enter radius of the circle: 4
Area of the circle:  50.24  squnit
Circumference of the circle:  25.12  unit
```

```
#Mean, Median and Mode.
import pandas as pd
d={'Name':pd.Series(['Tom','James','Ricky','Vin','Steve','Smith','Jack','Lee','Sai','Gaspe
df = pd.DataFrame(d)
print("Mean value in the distribution")
print(df.mean())#Calculating Mean
print("******************************")
print("Median values in the distribution")
print(df.median())#Calculating Median
print(df.mode())#Calculating Mode
```

```
Mean value in the distribution
Age       32.636364
Rating     3.743333
```

```
dtype: float64
******************************
Median values in the distribution
Age       30.00
Rating     3.79
dtype: float64
       Name   Age  Rating
0   Andreas  25.0    2.56
1    Gasper  30.0    2.98
2      Jack   NaN    3.20
3     James   NaN    3.24
4       Lee   NaN    3.65
5    Naviya   NaN    3.78
6     Ricky   NaN    3.80
7       Sai   NaN    3.98
8     Smith   NaN    4.10
9     Steve   NaN    4.23
10      Tom   NaN    4.60
11      Vin   NaN    4.80
<ipython-input-32-472f8c9a0ad4>:6: FutureWarning: The default value of numeric_
  print(df.mean())#Calculating Mean
<ipython-input-32-472f8c9a0ad4>:9: FutureWarning: The default value of numeric_
  print(df.median())#Calculating Median
```

```
#Correlation
import matplotlib.pyplot as plt
import seaborn as sns
df = sns.load_dataset('iris')
#without regresion
sns.pairplot(df,kind="scatter")
plt.show()
```

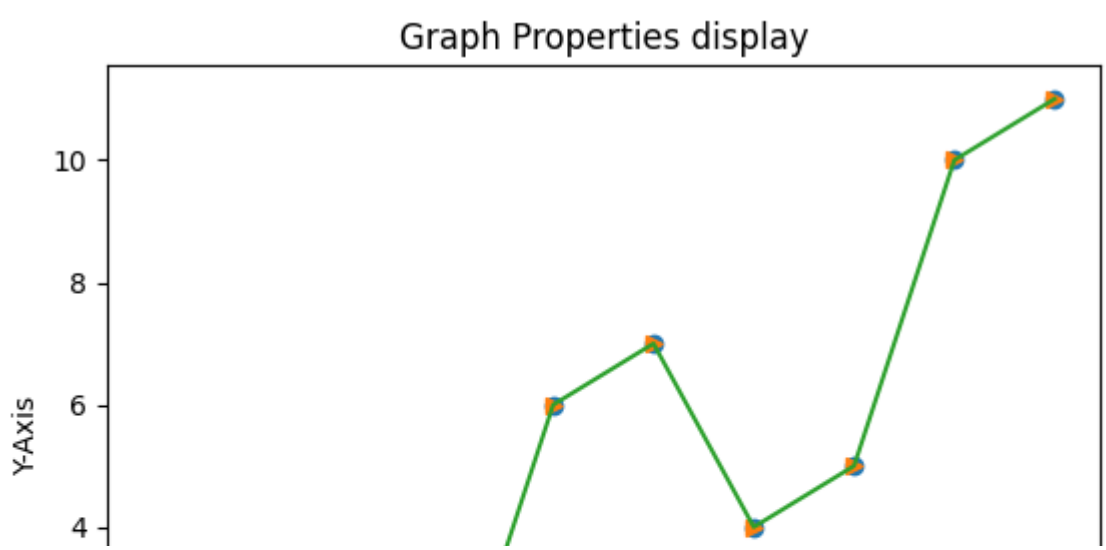



```
#Variance.
import pandas as pd
d={'Name':pd.Series(['Tom','James','Ricky','Vin','Steve','Smith','Jack','Lee','Sai','Gaspe
df = pd.DataFrame(d)
```

```
print(df.std())
print("**************")
print(df.skew())
```

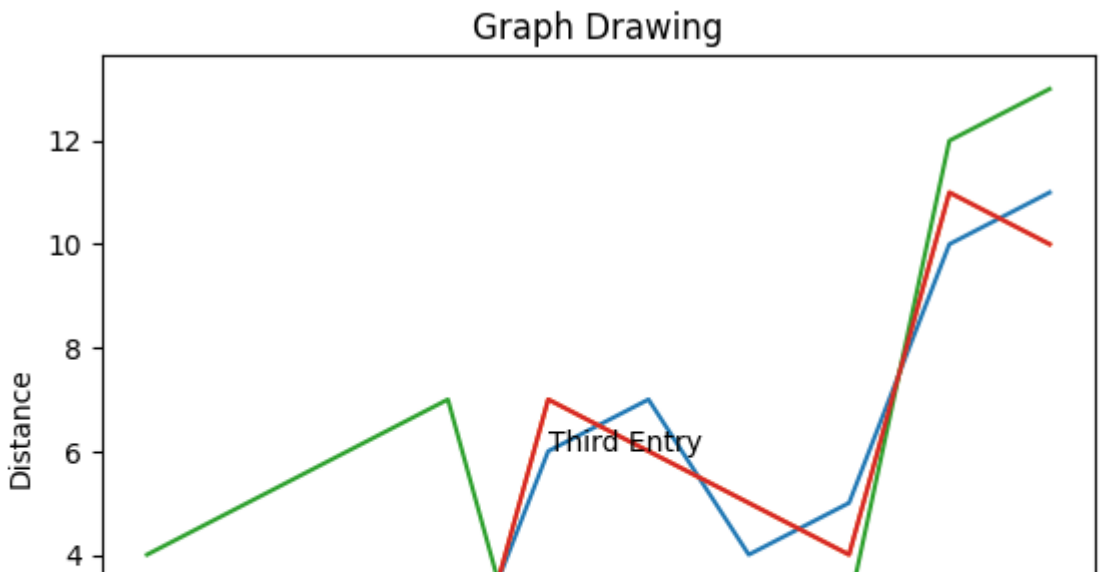
```
Age       9.233339
Rating    0.661628
dtype: float64
**************
Age       1.058687
Rating   -0.153629
dtype: float64
<ipython-input-5-fc0094163c75>:5: FutureWarning: The default value of numeric_o
  print(df.std())
<ipython-input-5-fc0094163c75>:7: FutureWarning: The default value of numeric_o
  print(df.skew())
```

```
#Chart Properties
import numpy as np
import matplotlib.pyplot as plt
x=np.arange(0,10)
y=x^2
#labeling the axes and title
plt.title("Graph Properties display")
plt.xlabel("X-Axis")
plt.ylabel("Y-Axis")
#formatting line colour
plt.plot(x,y,'o')
#formatting the line type
plt.plot(x,y,'>')
#simple plot
plt.plot(x,y)
#save as PDF
plt.savefig("Graph Properties Display in Python.pdf",format='pdf')
```




```
#keep this code line for graph display
```

```
import numpy as np
from matplotlib import pyplot as plt
x=np.arange(0,10)
```

```
y=x^2
z=x^3
t=x^4
#Labeling the axes and title
plt.title("Graph Drawing")
plt.xlabel("Time")
plt.ylabel("Distance")
plt.plot(x,y)
#Annonate
plt.annotate('Second Entry',xy=[2,1])
plt.annotate('Third Entry',xy=[4,6])
#Adding Legends
plt.plot(x,z)
plt.plot(x,t)
plt.legend(['Race1','Race2','Race3'],loc=4)
#Style the background
plt.style.use('fast')
```

```
[<matplotlib.lines.Line2D at 0x7c07ac0e7340>]
```